

Island biogeography of host-helminth interaction networks

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i Abstract

2 Introduction

3 Island biogeography theory posits that island area and distance to the mainland
4 combine to constrain species richness on islands, through effects on extinction and
5 colonization, respectively (MacArthur and Wilson, 2001). Despite being develop-
6 ed well over fifty years ago, it maintains an active research area, with numerous
7 open questions (Patiño et al., 2017). This is perhaps a result of the ways that
8 island biogeography theory can be used to inform other ecological relationships.
9 For instance, island biogeography theory is quite closely linked to the examination
10 of species-area relationships (Losos and Schluter, 2000; Peay et al., 2007). Further,
11 island biogeography theory is a simple base on which to build on, as seen by ex-
12 tensions to species functional trait variation (Jacquet et al., 2017), predator-prey
13 interactions (Gravel et al., 2011), and compositional differences between communi-
14 ties (Lu et al., 2018). It is clear that island biogeography theory has the potential
15 to inform trophic and community scale phenomena.

16 The increasing availability of ecological data has encouraged the study of large-
17 scale processes such as island biogeography theory. This data availability has also
18 promoted the study of species interaction networks (Delmas et al., 2019; Dormann
19 et al., 2017), which characterize the interactions between two (or more) trophic
20 groups (e.g., plant species and their pollinators). This has permitted examina-
21 tions into how spatial and environmental gradients influence the structure of these
22 species interaction networks (Tylianakis and Morris, 2017), with structure defined
23 using measures such as nestedness and modularity (Dalsgaard et al., 2013; Dor-
24 mann et al., 2017), which capture the tendency of species interactions to form
25 proper subsets of species with more interactions (nestedness) or cliques of inter-
26 acting species (modularity). Regardless of the measure used to capture aspects
27 of species interactions, there is mounting evidence that species interactions follow

the same patterns as species richness with regards to island area and isolation (Sugiura, 2010; Dormann et al., 2017; Tylianakis and Morris, 2017). For instance, Traveset et al. (2016) compared mainland and island plant-pollinator networks, finding that isolated oceanic islands had the lowest species richness – in agreement with predictions from classic island biogeography theory – and the lowest interaction diversity and highest plant niche overlap. That is, as the richness of plants and pollinators decreased with isolation, so did the structure of the interactions between plant and pollinator species. Relatedly, Dalsgaard et al. (2013) found that species interaction networks tended to be modular on islands, further linking island biogeography with the study of species interaction networks. But how do we interpret the effects of island isolation and area on species interactions?

Classical biogeography studies posit that the species richness of oceanic islands is largely determined by the equilibrium between species colonization and extinction rates. Island closer to mainland sources are colonized by new species at a higher rate, while islands with larger areas exhibit lower extinction rates. However, the interpretation of the influence of island isolation and area on the structure of species interaction networks must be different. That is, the richness of interacting species, even if driven by a balance between colonization and extinction rates, should not necessarily influence the overall structure of species interactions. For instance, if species interactions are conserved between mainland and island populations – as evidenced by Carstensen et al. (2018) – then differences between mainland and island species interaction networks should be minimal. However, there are other reasons why island area could influence network structure. Larger islands tend to have more diverse habitat types than smaller islands, and species habitat preferences can contribute to modularity of interaction networks by reducing niche overlap between groups of habitat specialists (Santos et al., 2016). Meanwhile,

island isolation, as noted above, is likely unrelated to network structure, provided that species interactions are not overly plastic (i.e., species can't readily change the set of species they interact with), or that species dispersal ability is directly related to specialization (i.e., good dispersers interact with many species).

It may be infeasible to provide tests of the underlying mechanisms which drive relationships between island isolation or area and properties of species interaction networks. However, demonstrations of the existence of the scaling relationships themselves are useful in identifying groups of species or types of interactions which modify the strength or directionality of effect of island characteristics on species interactions. For instance, while studies examining mutualistic network structure among islands are becoming increasingly common (Traveset et al., 2016; Dalsgaard et al., 2013; Sugiura, 2010; Trøjelsgaard et al., 2013), comparable studies of antagonistic networks have lagged behind (but see (Gravel et al., 2011)). However, antagonistic and mutualistic networks often have quite different structural properties, potentially owing to the intrinsic difference in the effect of the species interaction (i.e., the cost of parasitism and the benefits of mutualism). Here, we examine the relationship between host-parasite network structure and island isolation and size, using host-helminth occurrence data from 61 islands. First, we examined the relationship between species richness and island isolation and size, in accordance with traditional island biogeography theory. Second, we examined how a key measure of ecological network structure, interaction nestedness, scales with island characteristics. Overall, our findings suggest that while the diversity of host-parasite assemblages vary predictably with island characteristics, but the overall structure of their interaction networks do not. Reconciling the differences between mutualistic and antagonistic network structures among islands may provide insight into how island isolation and size are mechanistically related to species

80 interactions and network structure.

81 **Methods**

82 **Host - helminth interaction data**

83 Records of helminth parasite occurrence on host species were obtained from the
84 parasite database of the London Natural History Museum (LMNH) (Gibson et al.,
85 2005), and accessed programmatically using the `helminthR` package (Dallas, 2016).
86 These data currently represent one of the largest sources of host-parasite inter-
87 action data (Dallas et al., 2018), despite being restricted to helminth parasites,
88 including Platyhelminthes (trematodes and cestodes), Acanthocephalans, and Ne-
89 matodes (Gibson et al., 2005). We started by identifying a total of 62 islands for
90 which host-helminth interaction data were recorded (Figure 1). After filtering out
91 host and helminth species not identified to the species level, this comprised a total
92 of 7294 host-helminth species pairs across 2651 helminth species and 1980 host
93 species.

94 **Island isolation and size measurement**

95 Data characterizing the geography of islands were retrieved from Weigelt et al.
96 (2013), including island area (km²). We measure island isolation through distance
97 to the nearest mainland (km), and surrounding landmass proportion (SLMP),
98 calculated as the proportion of land area within 100, 1000, and 10,000km buffers.
99 This metric accounts for both spatial variation in mainland connection as well as
100 nearby island connections, and has been shown to be more robust than distance
101 per say in explaining insular plant diversity (Weigelt and Kreft, 2013). To roughly
102 account for historical connectivity, we also extract whether or not an island had
103 terrestrial connections to the mainland during the last glacial maximum (GMMC).

104 Records within the LNHM dataset were georeferenced to an island chain or series
105 (e.g. Trinidad & Tobago) were assigned to the largest island within the group (e.g.
106 Trinidad). Data from (Weigelt et al., 2013) were available for all but one island
107 in our data (South Shetlands), an Anartic island with only seabird and pinniped
108 infection records, which we excluded from our analysis.

109 **Quantifying host-helminth network structure**

110 We related island characteristics to both species richness estimates and to mea-
111 sures of interaction network structure. Measures of species richness for both hosts
112 and helminths were obtained from the London Natural History Museum data, as
113 described above. To quantify network structure, we estimated interaction nest-
114 edness, a commonly measured aspect of network structure found to be related to
115 network structural stability (Thébault and Fontaine, 2010; Dallas and Cornelius,
116 2015; Valdovinos, 2019) and resilience (Thébault and Fontaine, 2010; Suweis et al.,
117 2013). However, some measures of nestedness may be related to network size (num-
118 ber of interacting species)(Bascompte et al., 2003; Nielsen and Bascompte, 2007),
119 which creates a odd circularity where a significant effect of island area or isolation
120 on species richness would automatically manifest in an effect on network structure.

121 One method for addressing the difficulty in comparing network structure mea-
122 sures between networks is to measure the difference between the empirical statistic
123 and a distribution of the statistic obtained from null model randomizations. This
124 procedure has recently been criticized as not removing the effects of network size
125 (Chagnon, 2015), specifically for nestedness (Strona and Fattorini, 2014; Song
126 et al., 2017). Instead, we opt to use $NODF_c$, where normalized measure of the
127 classic nestedness metric based on overlap and decreasing fill proposed by Song
128 et al. (2017). Calculated as $NODF_c = NODF_n / (C \cdot \log(S))$, where $NODF_n$
129 represents the relative observed connectance of a given graph given its maximum

130 possible value ($NODF/\max(NODF)$), and C and S represent the connectance
 131 and the geometric mean of the number of nodes of each class, respectively. Song
 132 et al. (2017) show that this normalized metric is insensitive to network size, and
 133 thus appropriate for comparing network structure across biogeographic scales.
 134 Computation of the $NODF_c$ was performed using hill-climbing algorithm as im-
 135 plemented in the *maxNODF* package Hoeppeke and Simmons (2021). For some
 136 networks with a small number of interacting species, the set of possible network
 137 configurations is too small to making meaningful inference; as such we excluded
 138 islands with five or less species within their networks ($n=12$). Across the remain-
 139 ing 47 islands, most contained networks with multiple disconnected components
 140 (modules of species with no direct linkages that connected them, $n=35$). As $NODF$
 141 is not directly measurable in disconnected networks, for networks containing mul-
 142 tiple connected components we added the minimum number of links necessary to
 143 create a single component when calculating the maximum potential $NODF$ value
 144 (Hoeppeke and Simmons, 2021). We demonstrate in the Supplemental Materials
 145 how the use of network structure measures that contain the influence of network
 146 size would lead to incorrect conclusions regarding the influence of island area and
 147 isolation on network structure

148 **Statistical Analysis** Relationships between species richness and island char-
 149 acteristics were modeled as multivariate binomial regressions as implemented in
 150 the MASS package (Venables and Ripley, 2002). We It says in the form that
 151 landowners are not liable. Clayutilized binomial regressions as both poisson and
 152 quasipoisson regression models exhibited significant overdispersion (see supplemen-
 153 t). As our alternative metrics of isolation (log distance to mainland and SLMP)
 154 were significantly negatively correlated ($\rho = -0.768, p = 1.94e - 12$), we did not
 155 include both in the same model and instead used AIC to compare across metrics.

156 The impact of island characteristics on the spectral radius of its interaction metric
157 was modeled as a multivariate linear model using a gaussian error distribution
158 and logit link. Model fits were inspected through analysis of deviance and visual
159 inspect of standard model diagnostic visualizations. Models of the effect on histor-
160 ical mainland connectivity (GMCC) on diversity and network characteristics were
161 fit separately under the same framework as described above, as well as through a
162 Wilcoxon signed-rank test.

163 Results

164 Species richness and island characteristics

165 Island biogeography theory predicts that species richness will be negatively related
166 to distance from a mainland (as a result of low dispersal success to distant islands),
167 and positively related to island area (larger islands support more "stable" popula-
168 tions). We find evidence for an influence of area on species richness of both hosts
169 ($\beta = 0.419$, $p = 2.43\text{e-}12$) and helminths ($\beta = 0.391$, $p < 1.42\text{e-}10$). Interestingly,
170 host and helminth richness increased at nearly the same rate with island area. In
171 contrast to the influence of island area, we failed to detect any influence of distance
172 to the mainland on host ($\rho = -0.189$, $p = 0.182$) or helminth ($\beta = -0.221$, $p =$
173 0.116) species richness (Figure 2). However, when accounting for island connectiv-
174 ity through SLMP, we detected a positive relationship between surrounding land
175 area and richness of both hosts ($\beta = 1.54$, $p = 0.00161$) and helminths ($\beta = 1.74$,
176 $p = 0.000262$).

177 Network structure and island characteristics

178 Many measures of nestedness are sensitive to the effects of network size. In this
179 case, network size is the species richness of host and parasite species in the network,

180 which we found to be related to island area and SLMP (but not distance to the
181 mainland; Figure 2). Using a nestedness measure that was related to network size
182 would therefore lead to the conclusion that network structure follows a similar
183 gradient as species richness (see Supplemental Material). Here, we use a measure
184 of nestedness that is insensitive to network size, NODFc. Using this measure,
185 we detect a positive relationship between island area and interaction nestedness;
186 large island are more significantly nested ($\beta=0.325$, $p=0.0473$, Figure 3). We fail
187 to detect any relationship between network nestedness and maximum elevation,
188 distance to mainland, or SLMP.

189 Prehistoric Connectivity

190 Independent of contemporary island area or distance to mainland, historical land-
191 bridge connections may alter species assemblages or network characteristics by
192 influencing dispersal processes. We find that islands that had terrestrial mainland
193 connections during the last glacial maxima tended to have higher host ($\beta=1.61$,
194 $p=9.58e-05$) and helminth diversity ($\beta=1.59$, $p=8.11e-05$, Figure 4). Despite
195 differences in richness, interaction networks did not differ in nestedness across
196 GMMC status ($p=0.71$).

197 Discussion

198 Overall findings emphasizing why important

199 Overall, the relationships we observe between island characteristics and the
200 species richness of both hosts and helminth parasites are broadly consistent with
201 predictions from classical island biogeography theory for free-living species, but
202 network nestedness shows a more limited pattern, only responding weakly to island
203 area and not to either measure of island connectivity. This suggests that island

characteristics shape the size of host-parasite assemblages more predictably than they shape the organization of interactions within those assemblages. The framework of island biogeography predicts higher species richness on larger islands, in part because larger areas can support larger populations and thus lower extinction rates (MacArthur and Wilson, 2001), as well as greater habitat heterogeneity that accommodates a wider diversity of species (Triantis et al., 2008; Hortal et al., 2009; MacDonald et al., 2018). Consistent with these expectations, we find a significant positive correlation between island area and the species richness of both hosts and helminths.

Classical theory also predicts reduced diversity on islands that are more isolated from mainland source pools due to lower colonization rates (MacArthur and Wilson, 2001). While we do not detect a direct effect of distance to the mainland on species richness, we do observe a significant positive relationship between species richness and island connectivity, as measured by surrounding landmass proportion (SLMP), for both hosts and helminth parasites. Although distance and connectivity are related metrics, SLMP incorporates both proximity and the size and configuration of surrounding landmasses (including large neighboring islands), and may therefore provide a more accurate proxy for colonization potential (Weigelt et al., 2013; Weigelt and Kreft, 2013; Barreto et al., 2021).

Species interaction networks on larger islands tend to be more nested than those on smaller islands, but this effect was generally weak ($R^2 = 0.086$). In diverse communities, patterns of asymmetric specialization (i.e. specialist parasites tend to interact with hosts with high parasite richness, while the parasite communities of hosts with few associated parasites are dominated by generalist species) results in interaction nestedness (Vázquez et al., 2005; Lima Jr et al., 2012). Significant interaction nestedness has been observed in a number of parasitic systems (Krasnov

et al., 2011; Lima Jr et al., 2012; Guerra et al., 2016). The positive signal of island area on interaction nestedness may in part be driven by the absence of specialist species on small, host-poor islands. Or alternatively,

However, this mechanism cannot be fully explored without a more complete knowledge of helminth host range than is currently available. We have incomplete knowledge of which taxa the majority of helminths are able to infect in nature. The characterization of helminth host range, through a combination of increased parasite surveillance and experimental infection, can help yield insights into the mechanisms determining parasite colonization and ultimately network structure across biogeographical gradients.

Antagonistic versus mutualistic networks

While some attention has been paid to the island biogeography of mutualistic networks, the processes underpinning them are fundamentally different. Among these are the role of co-colonization; helminth generally have much lower dispersal capacity than their free-living hosts. As such, the dispersal of parasites to new islands is largely dependent on patterns of host dispersal, a feature already well described in invasion literature. Parasites that have high prevalence of infection or broad host ranges may be more likely to colonize new islands. For example, parasites with sufficiently broad ranges may be able to disperse to new areas via transient hosts, even if those species do establish persistent populations on a given island (Ellis et al. (2023) give an example with avian malaria). Alternatively, specialist species with close host associations may be more likely to be retained during colonizations (Schatz and Park, 2023). Linkages between host range and colonization potential may induce a biotic filter that ultimately effects network structural properties. Understanding how colonization dynamics scale up to affect resulting species interaction networks across gradients are an outstanding question.

256 **Caveats (sampling bias, distance to mainland calculation, ...)**

257 The sampling of parasite communities is inherently difficult, and is limited by the
258 availabiltiy of survey effort and taxonomic expertise. These data are unlikely to
259 be full characterizations of the host-helminth communities of these islands. This
260 is especially the case for very small islands (due to limited sampling effort) or very
261 large ones (which are too specious to reliably sample). A more complete and/or
262 effort-standardized survey of both parasite and host communities can better allow
263 us to make inferences across biogeographical scales.

264 **Broad conclusions**

265 However, large-data repositories like the London Museum of Natural History still
266 provide the opportunity to investigate the structure of parasitic communities at
267 macroecological scales, and can motivate these further data collection efforts. To
268 our knowledge, ours is the first study looking the the structure of parasite networks
269 within an island biogeography context. The increasing availability of open species
270 interaction data spanning wide geographic, taxonomic, and temporal scales pro-
271 vides the opportunity to better understand large-scale determinnants of network
272 structural properties. By looking across geographic and envnrionmental gradients,
273 we gain insight into the assembly processes that determine the structure and com-
274 position of antagonistic communities across environments.

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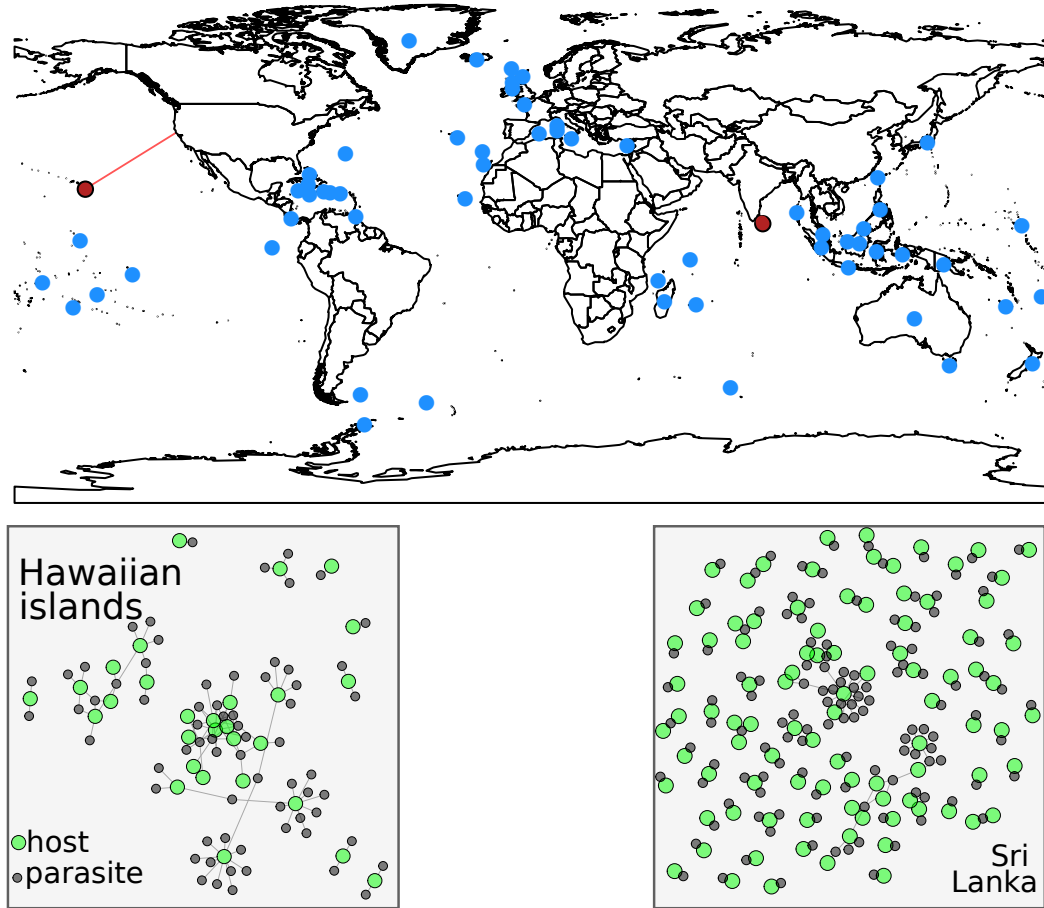


Figure 1: Both distance to mainland and island area may influence species richness and host-helminth network structure. Islands considered in the analysis are in blue, with two example networks from the Hawaiian islands and Sri Lanka highlighted in red. Hawaii is much smaller and more distant to the mainland, and also has lower species richness relative to Sri Lanka.

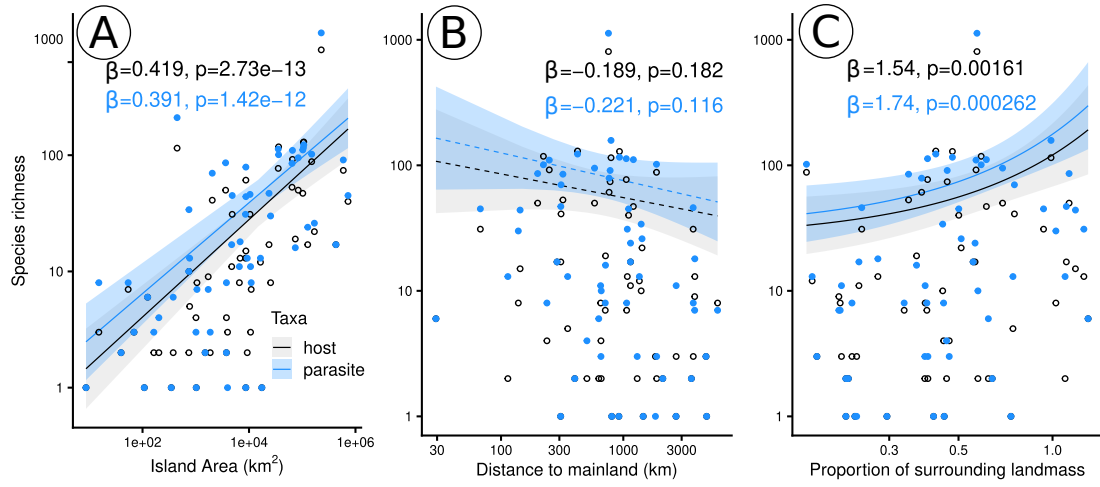


Figure 2: Traditional island biogeography theory relates species richness to the distance to the mainland (i.e., isolation) and island area. We detect a significant positive effect of island area on both helminth parasite and host richness (A), but found no significant effect of isolation as measured by distance to mainland (B). When measuring island isolation through surrounding landmass proportion, we did detect a weakly but significant positive effect of island connectivity on both helminth and host richness (C). Plotted lines represent fit relationships from a negative binomial multivariate regression model, assuming mean values of non-focal predictor variables. Significant relationships represented as solid lines; shaded regions represent 95% confidence intervals.

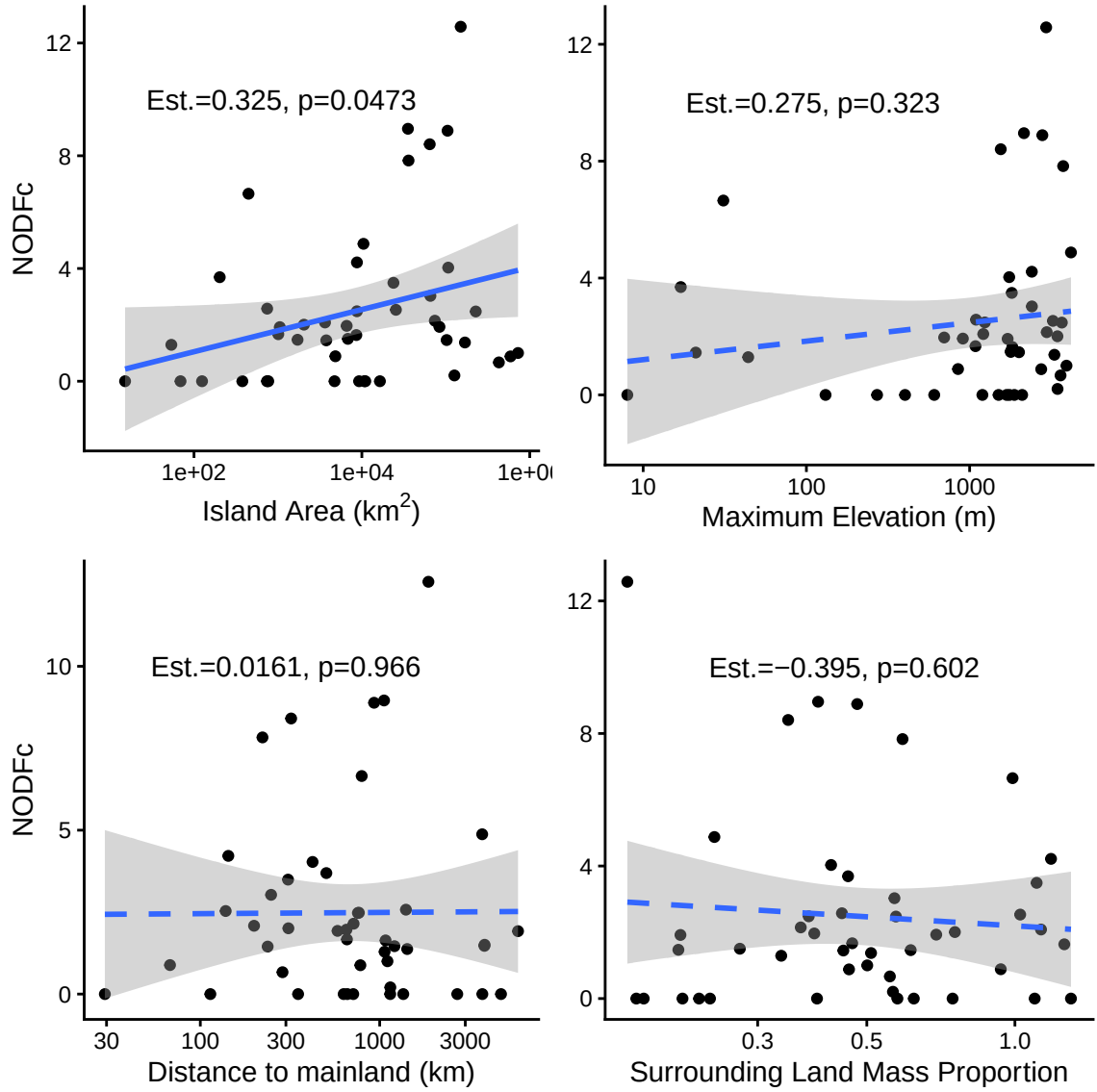


Figure 3: In addition to impacting network size, island area was positively related to $NODF_c$ (such that larger islands were tended to be more nested on average, top left). In contrast, neither island elevation (top right) nor measures of island isolation including distance from the mainland (bottom left) or SLMP (bottom right) were significantly related to interaction nestedness within island networks.

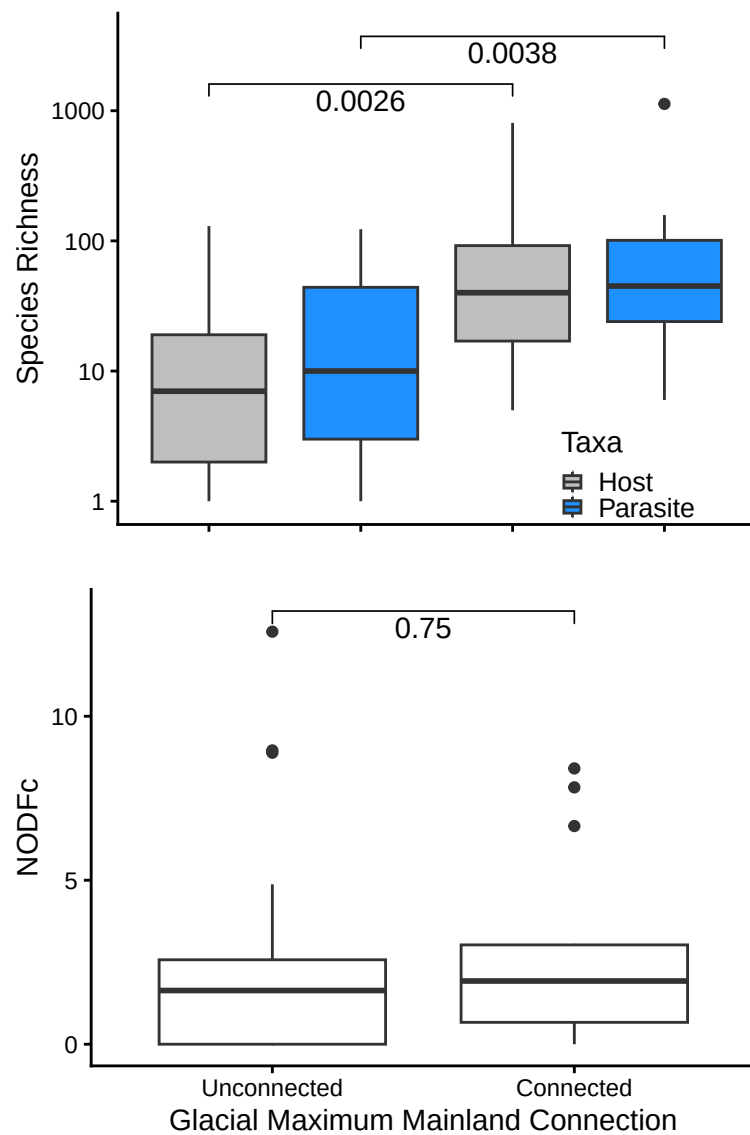


Figure 4: Islands with a terrestrial connection to the mainland during the most recent glacial maxima have significantly higher host and helminth parasite diversity (top), but exhibit no significant differences in network nestedness. Comparison labels indicate p-values from a pairwise Wilcoxon test.

Supplemental Material

Island biogeography of host-helminth interaction networks [a,*]Grant Foster [a]Tad A Dallas

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The relationship between network size and nestedness

Disentangling species richness (the sum of host and parasite richness) from nestedness is difficult, and many nestedness measures scale with species richness (Figure S1). Comparisons of properties across networks of different sizes is a non-trivial problem, and one potential solution is the implementation of a null modeling approach. In our case, we repeatedly shuffle links between hosts and helminths in each network, while maintaining both parasite host range (i.e., the number of host species a given parasite infects), and parasite species richness (i.e., the number of parasite species a host is infected by). For each network, we produced 1000 randomized networks, creating a distribution of spectral radii. Using these null distributions, we calculated z -scores, which measure the degree of divergence of the empirical network from the randomized null networks. Below, we present the correlations between alternative measures of network nestedness (η (Johnson et al., 2013), spectral radius, NODF, and $NODF_c$) as well as z -transformed versions of NODF and spectral radius. Nestedness as measured by η was not standardized, as null model randomization for this measure is difficult, as randomized networks maintaining aspects of the degree distribution of host and parasite species will result in the same η value as the empirical network.

We find a significant correlation between the log-scale number of species within a network and all size-alternative measures of network nestedness. However, as shown by Song et al. (2017), $NODF_c$ is unrelated to network size or connectance across large sets of randomly generated networks. The relationship we observe

436 across island networks may therefore reflect covariance with island area rather than
437 a direct effect of network size on nestedness. Island area is positively associated
438 with both network size 2 and network nestedness ??, suggesting that area may
439 act as a shared underlying driver that confounds the apparent association between
440 network size and nestedness. As such, using metric sensitive to network size may
441 give conflicting results. S3 illustrates that when using z-standardized NODF as
442 our measure of interaction nestedness, we see the opposite relationship between
443 island area and nestedness as with out $NODF_c$ metric, as well as relationships
444 with elevational range and SLMP. In contrast, measuring interaction nestedness
445 by z-standardized the spectral radius of a given graph yields qualitatively simliar
446 results to $NODF_c$ S3.

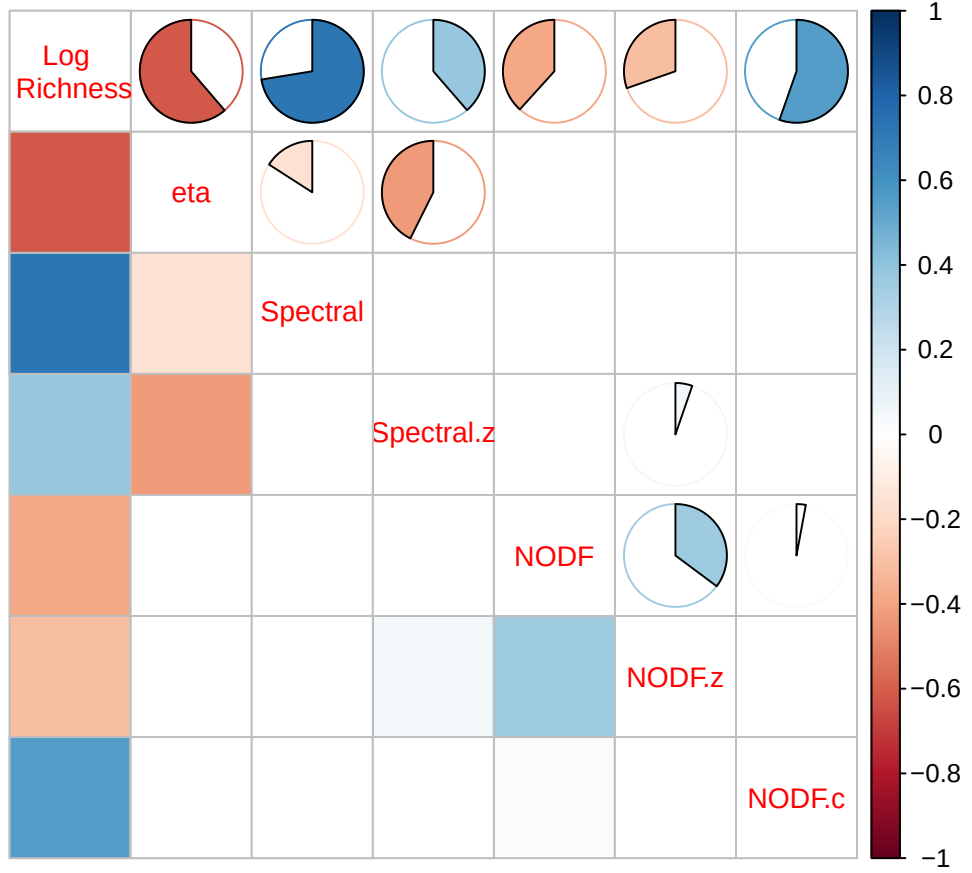


Figure S1: Correlogram visualizing pearson's correlation coefficients between network size (log species richness) and alternative measures of network nestedness in our data. Shaded squares represent significant correlations (p -value < 0.05). All measures of nestedness (η (Jonhson et al., 2013), spectral radius, and NODF, as well as z-scores of the latter two based on null randomizations) are significantly correlated with network size in our dataset

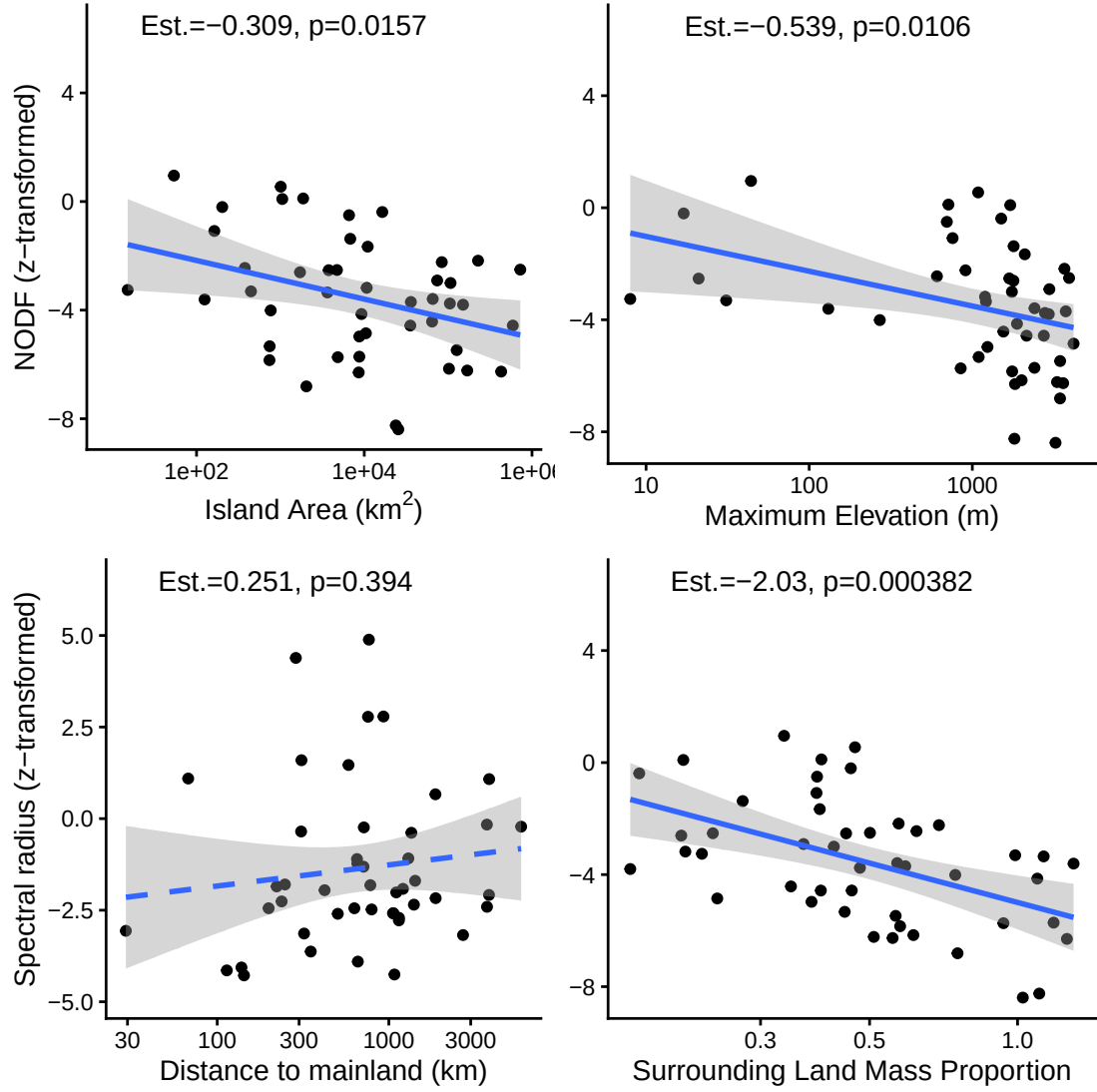


Figure S2: Recreation of Figure 3 of the main text using z-transformed NODF as our nestedness metric of comparison. Despite null modeling approaches, this metric may still be sensitive to network size, and this gives conflicting results to those observed in $NODF_c$.

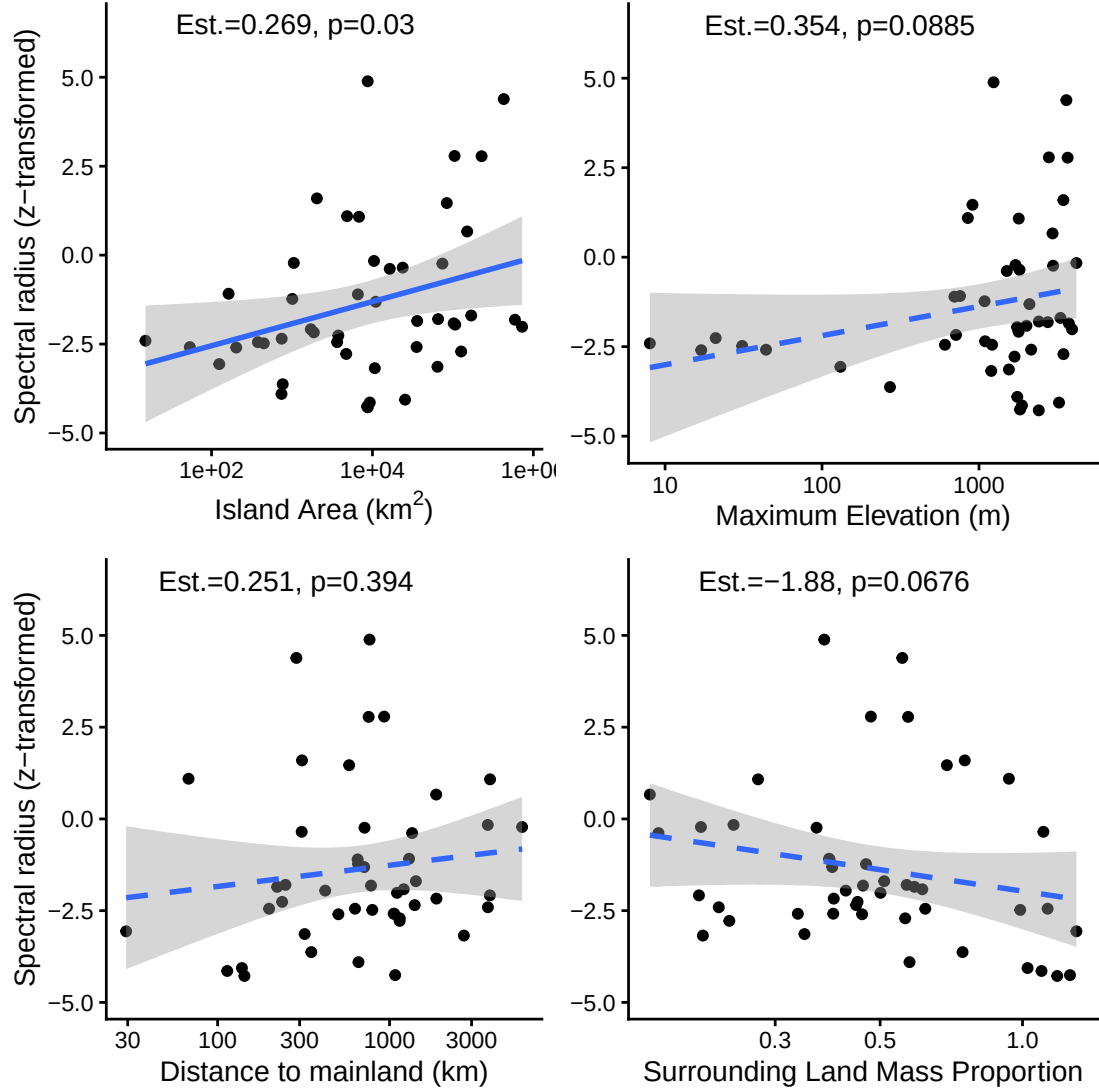


Figure S3: Recreation of Figure 3 of the main text using z-transformed spectral radius as our nestedness metric of comparison. In contrast to z-standardized NODF, spectral radius yields qualitatively similar results to $NODF_c$ after z-transformation.

447 Statistical Analysis

448 In the main text, we use a negative binomial modeling multivariate regression
 449 approach to model the effects of island characteristics on species diversity. This
 450 approach greatly minimizes residual deviance compared to both poisson and quasi-
 451 poisson glms $??$. Models incorporating island area and isolation as measured by
 452 SLMP yielded lowest AIC values, and as such are presented in the main text.

Table S1: Model deviance diagnostics for host diversity models.

Model	Residual deviance	Residual df	AIC	Deviance/d
Poisson: Area + SLMP	5621.91	55	5873.20	102.2
Quasi-Poisson: Area + SLMP	3497.06	55	–	63.5
Negative binomial: Area + Distance	65.10	55	488.86	1.1
Negative binomial: Area + SLMP	63.90	55	481.30	1.1
Negative binomial: Elevation + Distance	69.24	55	515.87	1.2
Negative binomial: Elevation + SLMP	68.08	55	509.48	1.2

453 Below, we report model diagnostic plots for the three models in the main text
 454 which show significant relationships between island characteristics and network
 455 properties.

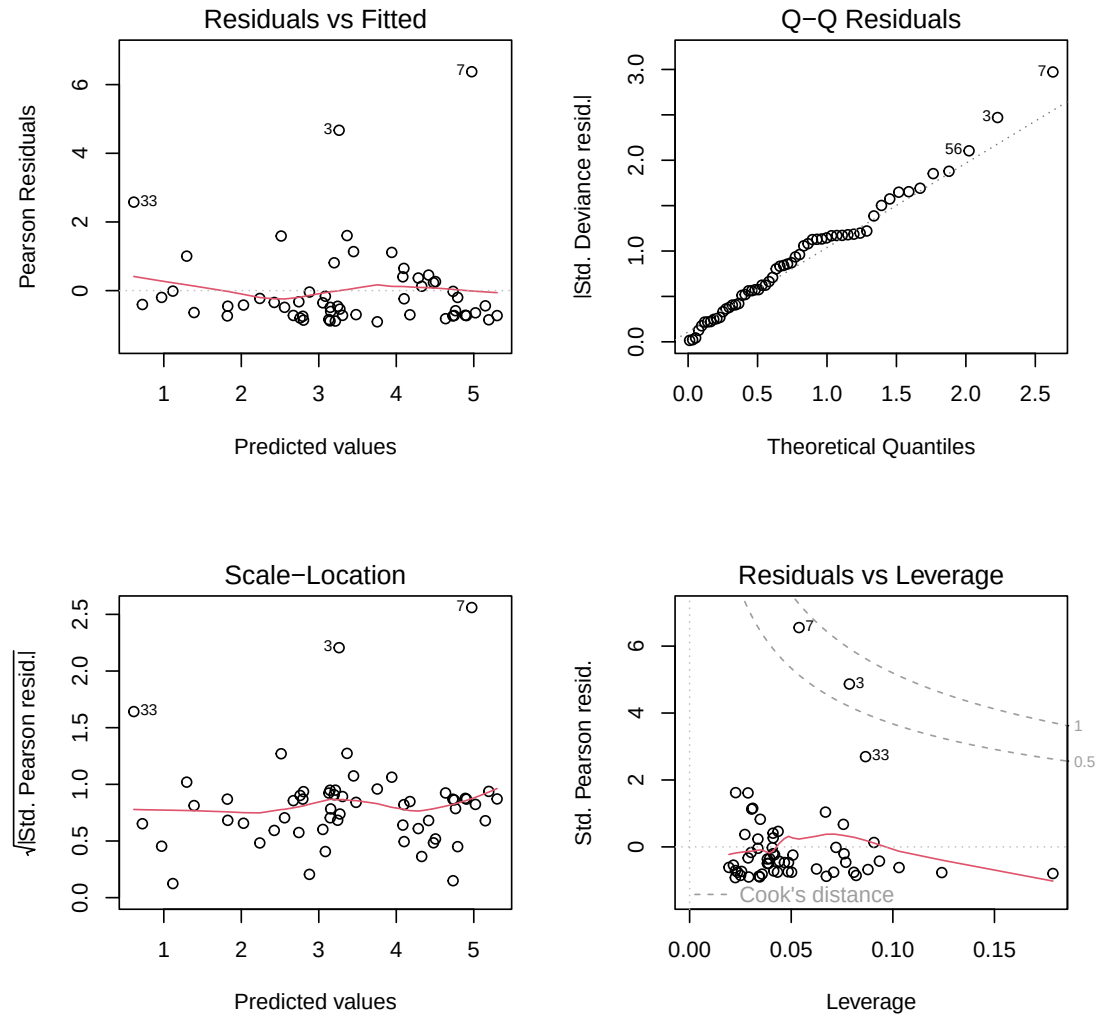


Figure S4: Model diagnostics plots for our negative binomial model of **parasite richness** across islands, including residual vs fitted values (top left), Q-Q plot (top right), scale-location (bottom left), and residual vs leverage plot (bottom right), with dotted lines representing Cook's distance contours of 0.5 and 1, respectively.

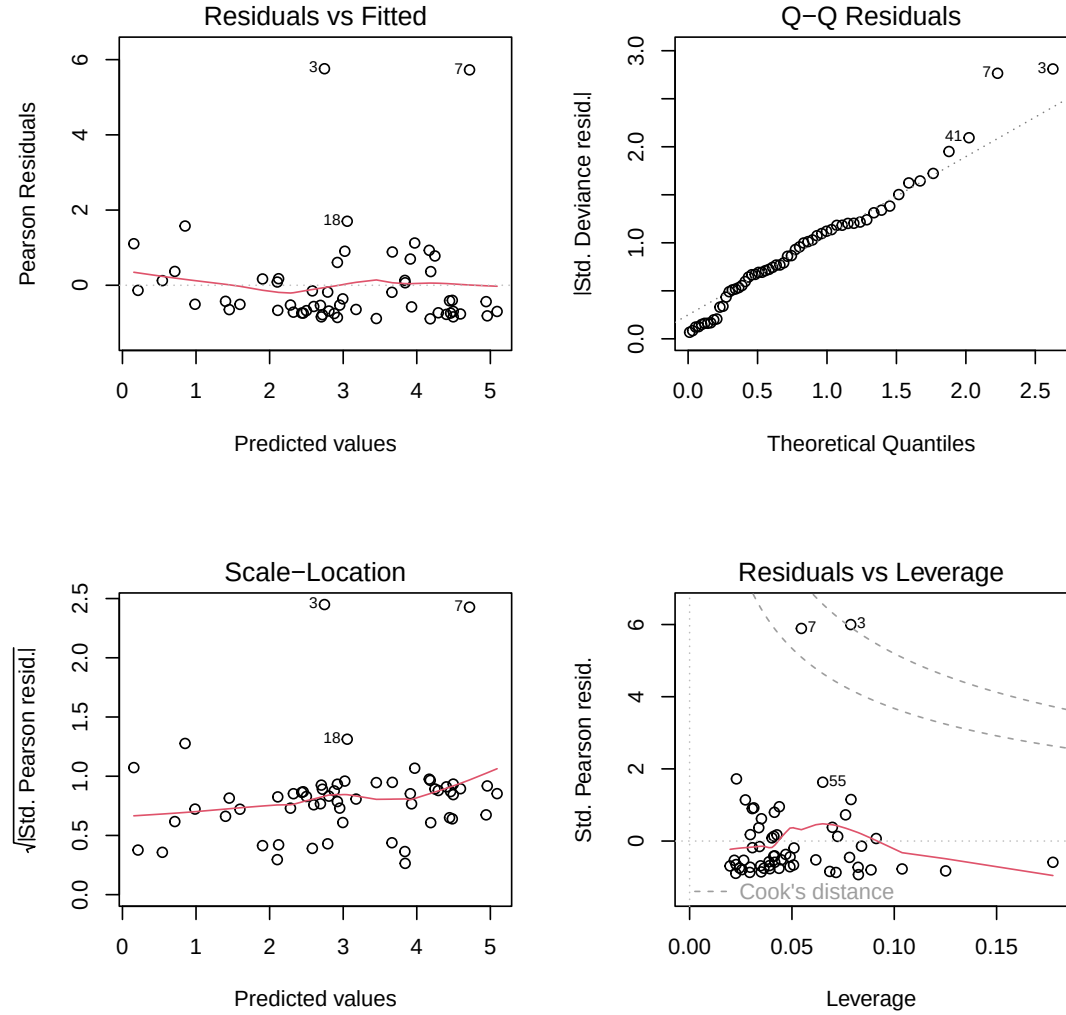


Figure S5: Model diagnostics plots for our negative binomial model of **host richness** across islands, including residual vs fitted values (top left), Q-Q plot (top right), scale-location (bottom left), and residual vs leverage plot (bottom right), with dotted lines representing Cook's distance contours of 0.5 and 1, respectively.

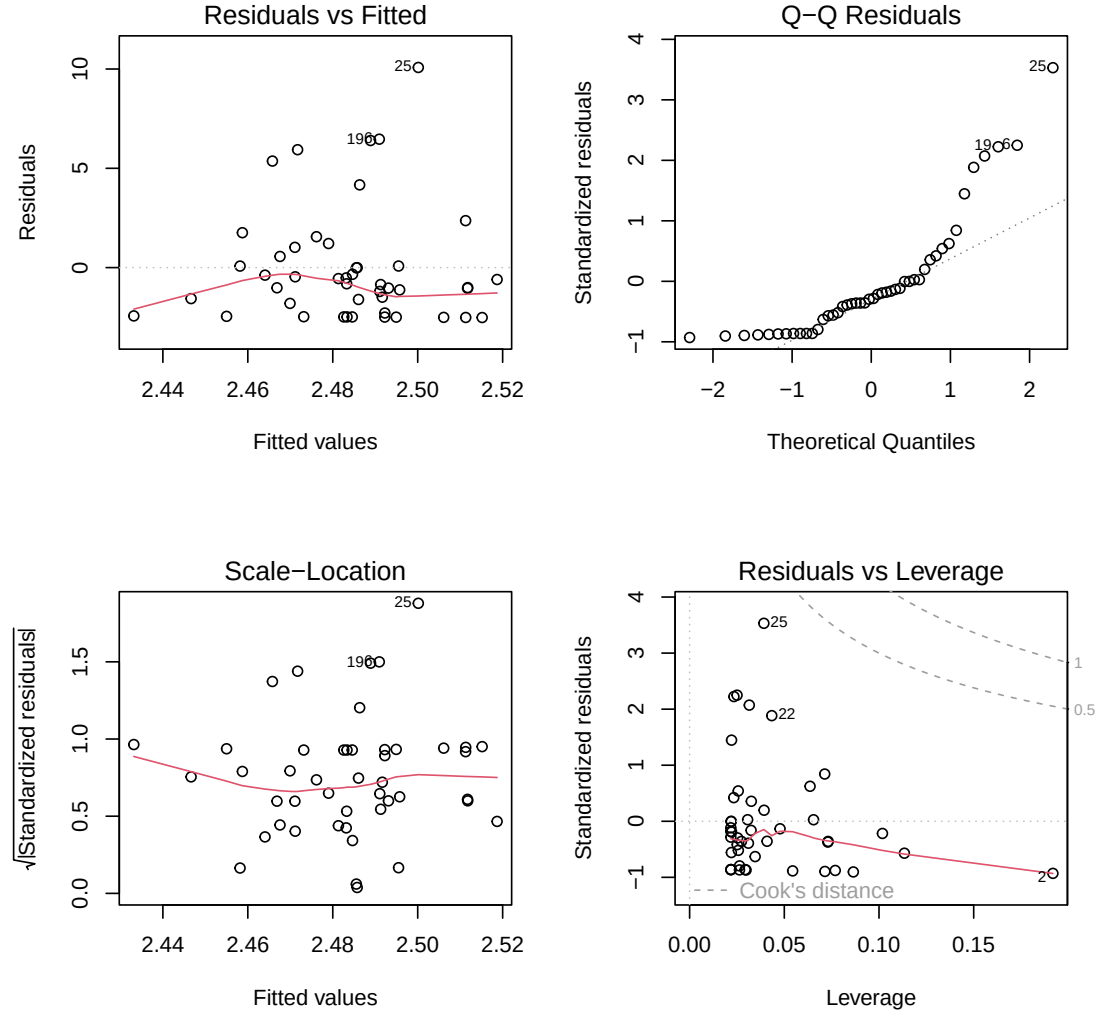


Figure S6: Model diagnostics plots for our generalized linear model of **network nestedness** ($NODF_c$) across islands, including residual vs fitted values (top left), Q-Q plot (top right), scale-location (bottom left), and residual vs leverage plot (bottom right), with dotted lines representing Cook's distance contours of 0.5 and 1, respectively.

456 **The relationship between nestedness and modularity**

457 We don't consider modularity in the main text, but we explore it here for fun, as
458 it is closely related to nestedness. Cite arxiv paper and Fortuna 2010 (?) and say
459 that they are like...the same thing.

460 GF: How necessary do you think this addition is? Can add it back in if you
461 think so, but would need to recalculate on our new network sets.